Olamilekan Matthew Omotosho

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Research Profile

AI researcher and engineer with academic and industry experience in machine learning, large language models, agent-based systems and AI evaluation. My work spans medical-image classification research, LLM fine-tuning and systematic evaluation, simulated environments for training and testing LLM agents, and benchmarking of tool-calling behaviour in language models. I want to study how autonomous agents can reason about the knowledge, beliefs and intentions of other agents, so that mixed teams of symbolic and learned agents coordinate reliably under uncertainty.

Research interests: epistemic planning; reasoning about knowledge and belief; multi-agent systems and coordination; knowledge representation and reasoning; decision-making under uncertainty; neuro-symbolic AI; evaluation and reliability of LLM-based agents.

Education

MSc Financial Engineering Jan 2025 – Present
WorldQuant University Cumulative average 88%

- 28 credits completed to date, including Machine Learning in Finance (93%), Deep Learning for Finance (95%), Stochastic Modeling and Financial Econometrics.

MBA, Specialisation in Artificial Intelligence Jan 2025 – Mar 2026
Nexford University CGPA 4.00/4.00

BTech Information Systems Feb 2019 – Oct 2024
Federal University of Technology, Akure (FUTA), Nigeria CGPA 4.83/5.00

- Best Graduating Student, School of Computing, and Best Graduating Student, Department of Information Systems.
- Dean's List in all 10 semesters.
- Thesis: *Ensemble Machine Learning Classification Algorithms for Diagnosis and Prediction of Lung Diseases.*

Research Experience

Research Collaborator, Low-Resource Function Calling Benchmark Oct 2026 – Present
MLC research focus group on Reinforcement Learning and Agentic Systems Remote

- Collaborating on *Can African Language Models Act? A Multi-Model Benchmark on Low-Resource Function Calling* (project lead: Ayodeji Lana), which tests how reliably African-language-focused LLMs select the correct function and construct correct arguments from requests in Yoruba, Hausa and Igbo.
- The design uses 200 single-call tasks drawn from the Berkeley Function Calling Leaderboard, each presented in the three languages and in matched English as a within-model control (800 inputs per model), with tool schemas and reference calls held fixed.
- Scoring separates structural validity, function selection, argument grounding and complete-call correctness, so that comprehension failures can be told apart from interface-value mismatches.
- Compares models adapted or built for African languages against general-purpose baselines to characterise failure patterns across languages and models.

Assistant AI Researcher Jan 2022 – Dec 2024
Federal University of Technology, Akure Nigeria

- Co-authored and co-presented a poster on efficient and fair CNN models for breast cancer diagnosis at the Women in Machine Learning (WiML) Workshop at NeurIPS 2024.
- Developed and evaluated medical computer vision models, reaching 93% accuracy in COVID-19 detection and F1 scores above 0.9 in chest X-ray disease classification.
- Contributed to a text-extraction model that identifies serial numbers on currency notes, combining text processing with automated information extraction.
- Reviewed machine learning literature and contributed to experimental design, model development and evaluation.

Undergraduate Thesis Research Completed Oct 2024
Federal University of Technology, Akure Nigeria

- Trained and compared CNN, Random Forest and XGBoost models on 11,985 chest X-ray images covering COVID-19, tuberculosis, pneumonia and normal cases.
- Combined the models in a stacking ensemble with a logistic regression meta-learner, reaching 96.77% classification accuracy.

Publications and Presentations

Workshop poster

- Akinrinwa, S. P., and **Omotosho, O. M.** (2024). *Towards Real-Time Breast Cancer Diagnosis: Enhancing CNN Model Efficiency and Fairness.* Poster, Women in Machine Learning (WiML) Workshop at NeurIPS 2024.

Research in progress

- *Can African Language Models Act? A Multi-Model Benchmark on Low-Resource Function Calling.* Collaborative project led by A. Lana; study design and evaluation under way.

Thesis

- **Omotosho, O. M.** (2024). *Ensemble Machine Learning Classification Algorithms for Diagnosis and Prediction of Lung Diseases.* BTech thesis, Federal University of Technology, Akure.

Professional Experience

Senior AI Engineer Jul 2026 – Present
Codematic Nigeria

- Design and build AI-powered applications that bring together language models, machine learning and software engineering.
- Develop LLM and agent workflows for production systems.

Pod Lead and Senior Data Scientist Apr 2024 – Present
Turing Remote

- Lead cross-functional teams of 3–6 data scientists in structured error analysis and comparative model evaluation to identify performance gaps in large language models.
- Built an automated LLM evaluation script that cut human review costs by 60% while keeping 90% agreement with expert raters.
- Design and refine supervised fine-tuning workflows that target hallucination, inconsistent reasoning and instruction-following failures.
- Evaluate model outputs for accuracy, reasoning, instruction following, harmlessness, style and verbosity to guide iterative improvement.
- Build simulated environments for training and evaluating LLM agents (RL environments): stateful mock APIs that mimic real-world services, in which a frontier model learns to plan and execute multi-step tool actions.
- Write scenario and regression tests for these environments and review agent trajectories to separate environment faults from model failures.
- Contribute to reinforcement learning from human feedback (RLHF) workflows and training-data curation.

Data Scientist Jun 2023 – Aug 2024
VAMP FI Hybrid

- Built a conversational agent with LangChain and the OpenAI API that delivers real-time insights from operational dashboards.
- Developed a prescriptive machine learning model for facility optimisation that reduced operational time by 20%.
- Used Python and SQL to validate data models and led data quality work to improve data integrity across the organisation.
- Worked with UX designers to turn analytical findings into usable dashboards and visualisations.

Data Science Trainer Jan 2024 – Apr 2024
Analytics Intelligence Hybrid

- Developed and delivered training in Python, machine learning and deep learning, with practical projects in computer vision, NLP and predictive analytics.

Data Analyst Jul 2022 – Aug 2024
Filmo Real Estate Remote

- Built Power BI dashboards that improved SLA adherence by 30%, and automated reporting with SQL and Power Query, cutting manual reporting work by 50%.
- Optimised database queries and workflows, reducing system load times by 87%.

Data Analyst Oct 2020 – May 2022
Quicktest Remote

- Developed an automated personalised messaging system in Python that increased customer retention by 55%.

Freelance AI/ML Engineer Jan 2020 – Present
Independent Remote

- Build LLM-powered applications and machine learning solutions in NLP, computer vision and predictive modelling; compete in Kaggle and Zindi machine learning competitions.

Selected Projects

Offline LLM Learning Platform Python, FastAPI, Ollama, Docker, Google Cloud Run

- Built an end-to-end application that generates curriculum-grounded lesson plans using Gemma 3 1B, curriculum documents and teacher profiles, served by a FastAPI backend deployed to Google Cloud Run.
- Added student intent classification and hallucination guardrails to keep generation grounded in context.

Build Your Own RAG System Python, Streamlit, vector search, Gemini, Ollama

- Built an interactive retrieval-augmented generation application covering document ingestion, vector search, semantic retrieval and conversational question answering grounded in source material.

Real-Time Analytics Agent Python, LangChain, OpenAI API

- Built an LLM agent at VAMP FI that converts analytical data into answers to operational questions and automated insights.

Air Quality Sensor Fault Classifier UmojaHack Africa 2022; Python, scikit-learn

- Developed a Random Forest sensor-fault classifier with time-series feature engineering, reaching 93% accuracy under stratified k-fold validation on more than 290,000 records.

Technical Skills

LLMs and agents	LangChain, OpenAI API, agent workflows and tool use, function calling, simulated agent (RL) environments, RAG
LLM training and evaluation	Supervised fine-tuning, RLHF workflows, benchmarking, automated evaluation, error analysis
Machine learning	PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, CNNs, ensemble methods
Programming and data	Python, SQL, Pandas, NumPy, FastAPI, REST APIs, data pipelines
Engineering and cloud	Git, Docker, Google Cloud, Azure, AWS, Ollama
Research tools	LaTeX, JupyterLab

Awards and Scholarships

- Top 100 Africa Future Leaders (2026).
- Best Graduating Student, School of Computing, Federal University of Technology, Akure (2024).
- Best Graduating Student, Department of Information Systems, Federal University of Technology, Akure (2024).
- Dean's List, 10 consecutive semesters (2019–2024).
- NNPC/Chevron Nigeria Limited Joint Venture National University Scholarship Award (2020).
- Recipient of five undergraduate scholarships in total, including the FSB scholarship.

Academic Service and Leadership

- Application reviewer, Deep Learning Indaba, University of Rwanda, Kigali (2025); delegate and reviewer (2022–2024).
- Zindi Ambassador, Zindi Africa (2023–Present).
- Community Lead, Data Science Nigeria, FUTA chapter (2023–2025).
- Mentor, 3 Million Technical Talent (3MTT) Programme, Nigeria (2026); trained and mentored more than 80 participants.
- Mentor, Young Data Professionals Mentorship Program (2026), and Ingressive For Good.
- Host and Publicity Lead, Microsoft Learn Student Ambassadors (2021–2024).

Certifications

- Google Cloud Certified Partner Specialist: Gemini Enterprise Agent Development (2026)
- Google Cloud Certified: Generative AI Leader (2026)
- Google Cloud Certified: Professional Machine Learning Engineer (2026)
- Microsoft Certified: Azure AI Engineer Associate (2025)
- Microsoft Certified: Azure Data Scientist Associate (2024)
- Microsoft Certified: Power BI Data Analyst Associate (2023)
- Microsoft Certified: Azure AI Fundamentals (2023)
- Microsoft Certified: Azure Data Fundamentals (2022)
- Udacity Nanodegree: Data Analyst (2022)
- Tekedia Institute: Advanced Diploma in Business Administration (2022)
- Google Data Analytics Professional Certificate (2022)
- DeepLearning.AI: Convolutional Neural Networks (2021)
- DeepLearning.AI: Improving Deep Neural Networks (2020)
- DeepLearning.AI: Neural Networks and Deep Learning (2020)